Semantic Information Gain for Robust Human-in-the-Loop Program Synthesis

A Partial Implementation and Controlled Evaluation

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1. Ambiguous specifications

Specification: "Remove duplicate elements."

- Stable first occurrences: `[4, 1, 2]`.
- Sorted unique values: `[1, 2, 4]`.

Both interpretations fit the description for input `[4, 1, 4, 2, 1]`.

SEMSYNTH interaction

Generate a candidate pool, select a distinguishing question, and revise the pool after a Yes/No answer.

The original design aims to reduce the effort of expressing intent. This project evaluates candidate selection; human effort has not been measured.

Source: Supplied SEMSYNTH manuscript; course report, Introduction and Method

2. Original SEMSYNTH architecture

1. **Specification parsing:** goals, constraints, and interface.
2. **Feature extraction:** 47 code, execution, and question features.
3. **AGA initialization:** generate strategies, then translate to code.
4. **Question selection:** semantic information gain and diversity.
5. **Oracle interaction:** atomic Yes/No clarification.
6. **Belief revision:** Bayesian updates and consistency checks.
7. **Termination / regeneration:** select a candidate or rebuild the pool.

Course scope: a bounded execution-based subset of this architecture.

Source: Supplied SEMSYNTH manuscript; course report, Method

3. Original feature model and question design

47-dimensional representation

- 18 AST structural features.
- 18 execution and semantic features.
- 11 question-text features.

XGBoost predicts satisfaction:

$$M(h, q) = P(h \models q).$$

Not trained or integrated in this prototype.

Semantic question design

- 85 templates in nine clusters.
- Atomic decomposition of compound queries.
- Diversity penalty on repeated aspects.

$$IG_{\text{final}}(q) = IG(q) - \lambda \max_{q' \in Q_{\text{hist}}} \text{Sim}(q, q').$$

Retained as design; not evaluated as implemented modules.

Source: Supplied SEMSYNTH manuscript; course report, Method

4. Information gain in the implemented special case

$$IG(q) = H(\mathcal{H}) - \sum_{a \in \{\mathrm{Yes},\mathrm{No}\}} P(a \mid q) H(\mathcal{H} \mid a, q).$$

Assumptions: uniform priors, deterministic probe execution, exact feedback.

For the question "On input x, should the output equal y?", let $n = |\mathcal{H}|$ and $p = |\mathcal{H}_{\mathrm{Yes}}|/n$. For nonempty parts,

$$\begin{aligned} IG(q) &= \log_2 n - p \log_2(pn) - (1-p)\log_2((1-p)n) \\ &= -p \log_2 p - (1-p)\log_2(1-p) = H_b(p). \end{aligned}$$

Meaning for the prototype

Balanced selection maximizes this score over available probe partitions. With three candidates, a 1–2 split is the closest available balance. Bounded probes can still leave candidates unresolved.

Source: Course report, Method: derivation and conditional query bound

5. Implementation scope

Component	Evaluated status
Candidate pool and execution	Adapters / fixed banks, admission, deduplication, isolated execution.
Question selection	Binary entropy over bounded I/O probe partitions.
Belief update	Hard filtering measured; optional soft updates have unit tests only.
Oracle and stopping	Noiseless reference oracle; single, empty, budget, or unresolved states.
UI and experiments	Streamlit, shared pools, checkpoints, CSV, replay.
Remaining original modules	47-feature mapping, full atomic/diversity, AGA, consistency recovery, hierarchy, and regeneration not integrated.

Source: Course report, Implementation Scope; current src/ and experiment configuration

6. Evaluation data and pilot provenance

Measured benchmark: 32 synthetic cases

- Eight related template families, four variants each.
- Dedup, sorting, intersection, parity, case, trim, threshold, palindrome.
- Offline: three candidates per bank, including a reference solution.
- Two saved LLM-labeled pilots: eight synthetic cases each.

Separate pilot catalog: 15 local tasks

- HumanEval / MBPP sources checked; 46 local tests pass.
- MBPP IDs corrected to 2 and 91 for two tasks.
- Uppercase count is local; literal search adapts MBPP/534.
- Local tests are reduced/adapted, not an official benchmark suite.

Source: config/; docs/dataset_catalog.csv; dataset_audit.json and synthetic_audit.json

7. Controlled evaluation protocol

Shared ordered pool per task

- Direct-1: first candidate, no questions.
- First-valid: first admitted candidate, no questions.
- Balanced: entropy-based questions and hard filtering.

Seed 3101, requested $K = 3$, at most three questions and eight probe inputs. Saved pilot pools contain two or three candidates.

Exact-input separation

$$X_{\text{eval,clean}} = X_{\text{eval}} \setminus X_{\text{probes}}.$$

- Offline: 33 of 160 input instances excluded; 127 retained.
- Each eight-case pilot: 31 retained input instances.
- Success means passing all retained tests for a case.

Reference feedback is noiseless. Input separation does not establish generalization to new task families.

Source: Saved run manifests/checkpoints; course report, Experimental Setup

8. Controlled offline results

Family	Direct-1	First-valid	Balanced	Mean Qs
Case	3/4	3/4	4/4	1.25
Dedup	2/4	2/4	4/4	1.50
Intersection	1/4	1/4	4/4	1.25
Palindrome	2/4	2/4	4/4	1.50
Parity	1/4	1/4	4/4	1.75
Sorting	1/4	1/4	4/4	1.75
Threshold	0/4	0/4	4/4	1.50
Trim	3/4	3/4	4/4	1.25
Overall	13/32	13/32	32/32	1.4688

- Balanced: **32/32** finite-suite case successes vs. **13/32** for each baseline.
- Difference: **+59.375 percentage points**; mean **1.4688 questions**.
- Reference is in every bank. This measures selection under controlled ambiguity.

Source: Offline run e6a84cf2098af752; table generated from saved checkpoints

9. Saved LLM-labeled pilot pools

Recorded identifier	Selector	Passed	Rate	Mean Qs
`gemini-3.5-flash-lite`	Direct-1	7/8	87.5%	0.00
`gemini-3.5-flash-lite`	First-valid	7/8	87.5%	0.00
`gemini-3.5-flash-lite`	Balanced	8/8	100.0%	0.25
`qwen2.5-coder:32b`	Direct-1	6/8	75.0%	0.00
`qwen2.5-coder:32b`	First-valid	6/8	75.0%	0.00
`qwen2.5-coder:32b`	Balanced	8/8	100.0%	0.50

- Eight synthetic cases per recorded identifier; common pool across selectors.
- Balanced gains: +12.5 and +25.0 percentage points over Direct-1.
- Checks replay saved pools. Original API calls, model weights, tokens, and generation latency are not independently reverified.

Source: Runs 8be202264a584167 and b6c988199ed7b1d5; saved manifests/checkpoints

10. Replayed clarification and stopping states

Qwen-labeled dedup case

Input: `[4, 1, 4, 2, 1]`.

- Candidate 1: `list(set(xs))`.
- Candidate 2: stable first occurrences.

Question: "Should the output equal `[1, 2, 4]`?"

Reference answer: **No**. One candidate remains, returning `[4, 1, 2]`; all four retained tests pass.

Balanced stopping states

Run	Single	Unresolved
Offline	29	3
Gemini	2	6
Qwen	3	5

Unresolved means no distinguishing question was found in the bounded probes. It does not prove functional equivalence.

A single survivor also does not prove correctness on all inputs.

Source: evidence_replay.json; SYN_DEDUP_01 and stopping counts from saved checkpoints

11. Conclusions and remaining original modules

Implemented and measured

- Execution-based binary clarification and hard filtering on small pools.
- 32/32 controlled offline successes with a reference in every bank.
- Saved pilot selection: 8/8 per identifier on retained finite test suites.

Results support this subset under the stated protocol. Human effort, noisy feedback, and broad generation accuracy remain unmeasured.

Original design retained for future integration

- Train and calibrate 47-feature $M(h, q)$.
- Integrate atomic questions and diversity.
- Add consistency recovery, two-stage AGA, and regeneration.
- Evaluate larger spaces and real users.

Thank you

Source: Course report, Discussion and Conclusion; supplied SEMSYNTH design